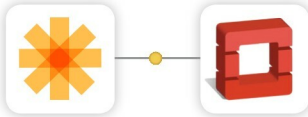


Open Daylight

SDN Integration with open stack



OpenDaylight is a collaborative open source project hosted by the Linux Foundation. The goal of this project is to accelerate the adoption of software defined networking (SDN), creating a solid foundation for network functions virtualisation (NFV). OpenStack is a FOSS software platform for cloud computing, generally deployed as Infrastructure-as-a-Service (IaaS). This article gives a step-by-step tutorial on integrating OpenDaylight and OpenStack.

The adoption of open source technologies is on the rise, which is evident from the popularity of projects like OpenStack and OpenDaylight. While each project tries to address specific problems, a combination of OpenStack and OpenDaylight projects might be something to look out for.

OpenDaylight started with the release of Hydrogen and is now in its third release cycle, named Lithium. There are more than 60 projects available or being incubated with OpenDaylight Lithium that could satisfy all the networking requirements demanded by SDN. There are also some dedicated projects for OpenStack that are designed and integrated with OpenDaylight.

In this article, I will walk you through the steps in deploying OpenDaylight and integrating it with OpenStack to address use cases specific to networking infrastructure.

1. Assumption

Before getting started, I assume the reader understands OpenStack (kilo) and the set-up is already done.

2. Installing OpenDaylight

OpenDaylight Controller runs in a JVM. So install OpenJDK7 using the command below:

```
$ apt-get install openjdk-7-jdk
```

Download the latest OpenDaylight-Lithium package from the official repo, as follows:

```
$ wget https://nexus.opendaylight.org/content/groups/public/org.opendaylight/integration/distribution-karaf-0.3.3-Lithium-SR3/distribution-karaf-0.3.3-Lithium-SR3.tar.gz
```

Uncompress it as the root user, and start OpenDaylight

using the following command:

```
$ tar xvfz distribution-karaf-0.3.3-Lithium-SR3.tar.gz
$ cd distribution-karaf-0.3.3-Lithium-SR3/
$ ./bin/start # Start OpenDaylight as a server process and
Wait for some time before running the next command below.
```

Connect to the Karaf shell, as follows:

```
$ ./bin/client # connecting to the OpenDaylight with the
client
```

Now, you should be in OpenDaylight's console. Install all the required features:

```
opendaylight-user@root> feature:install odl-base-all odl-aaa-
authn odl-restconf odl-nsf-all
odl-adsal-northbound odl-mdsal-apidocs odl-ovsdb-openstack
odl-ovsdb-northbound odlidlux-core
#Feature installation may take some time to install.
```

Once the installation is complete, you can check whether everything is working fine by using the *curl* call shown below:

```
$ curl -u admin:admin http://<OPENDAYLIGHT SERVER's IP>:8080/
controller/nb/v2/neutron/networks
```

The response should be an empty network list if OpenDaylight is working properly.

Also, you should be able to log in to the *dlux* interface on <http://<OPENDAYLIGHT SERVER's IP>:8181/index.html> - the default username and password is 'admin/admin' (see Figure 2).

Additionally, you can find the log details at the location below: